LDMS USER GROUP MEETING

May 4, 2026

Agenda

- LDMSCON2026 – Reminder
- LDMS Multi-Tenant Security Working Group - Update
- V4.5.2
- Q&A

LDMSCON2026

LDMSCON2026

https://sites.google.com/view/ldmscon2026/home

- Reminder to submit your abstracts
  - Extended deadline to May 11 (Next Monday)
  - https://easychair.org/conferences/?conf=ldmscon2026
- Reminder to register
  - Hard deadline: May 15 (Next Friday)
  - https://ldmscon2026.eventbrite.com/

OGC

Sandia National Laboratories

LDMS Multi-Tenant Security Working Group — Update

Multi-Tenant Security WG — Second Meeting Recap

Part 1: Existing setup

- Current model runs ldmsd on host as root with trust between daemons
  - or as user with `ovis` auth
- Sets are created by trusted ldmsd
- Messages are authenticated at the entry point via munge, with propagation controlled by ldmsd configuration
- **Recommended setup if ldmsd run as non-root:**
  - Munge on localhost for local connections (ldmsd and other apps)
  - ovis auth for remote — with an optional exporter layer to control what gets exported to whom
- **Limitations**: UID/GID assumes a single user realm across all nodes, and there is currently no restriction on who can publish messages

Multi-Tenant Security WG — Second Meeting Recap (cont.)

Part 2: Certificate-Based Auth (Exploration)

- **ED25519 certificate is a possible approach** — 128-byte certs that don't rely on local UID/GID, usable for transport authentication and for signing messages or sets
- **Signing sets vs. signing messages**: both add 192 bytes overhead; signing sets verifies the creator, signing messages verifies user identity and message authenticity end-to-end
  - signing adds CPU overhead on publisher (usually compute nodes)
  - verifying adds CPU overhead on consumers and intermediate processes
- **Open questions that were discussed**:
  - who manages user certificates instead of relying on Linux UID/GID
  - whether Linux UIDs are consistent across clusters at a site

Multi-Tenant Security WG — Second Meeting Recap (cont.)

Part 3: Munge + Container Finding

- Containers (Podman - user namespace isolation)
  - Started by user UID 1000
  - Mounted /run/munge
- Run as root inside container:
  - Munged on host sees host UID, e.g., 1000
- Run as non-root inside container:
  - Munged on host sees user’s subUID, e.g., 100007

V4.5.2

V4.5.2 – Changes

<u>Update configs</u> –

- “message_channel” → “message_tag” in prdcr_subscribe/prdcr_unsubscribe
  - “message_channel” is an alias now → You don’t need to change your config files now
  - “message_tag” is preferred

YAML Parser

- Hostlist now numerically sorted
- Various bug fixes and improvements

V4.5.2 – Changes (cont.)

LDMS Transport

- RDMA fixed – backward compatible with v4.5.1
  - LDMS RDMA on V4.5.1 doesn't work on OmniPath and IB

LDMS Messages

- Race fix in CLIENT_CLOSE
- `setns()`/fork conflict with `slurm_notifier` (LDMS Spank Plugin)

V4.5.2 – Changes (cont.)

Sampler Plugins

- **slurm_notifier**: bad library linkage fixed (immediate-binding systems)
- **filesingle**: checks both cpuinfo_cur_freq and scaling_cur_freq
- **lustre samplers**: internal refactor — per-instance state, sanitized job_id strings
- **slurm_sampler2**: all-tasks-exit no longer implies job exit, use the job exit event as a signal

Storage Plugins

- **store_influx**: no longer crashes on sets without job_id

Decomposition

- **decomp_static**: group timeout parsing fixed

Q & A

- Issue 2022 – Leak in the config handler (May not need to hold up for v4.5.2)

Thank you!